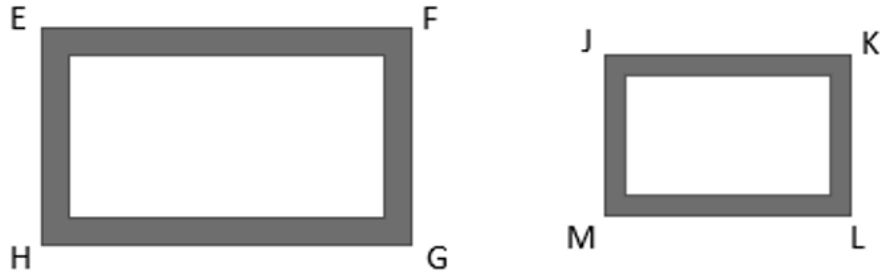


Geometry
Unit 6
Lesson 12 Practice

Name _____
Date _____
Period _____

Tamika is custom building frames for a set of family photos. She has designed two similar rectangular frames, $EFGH$ and $JKLM$. The diagram shows the two different sized frames, where $JM = 6$ inches (in.), $EF = (3w + 2)$ in., $FG = 10$ in., and $JK = (2w + 0.8)$ in.



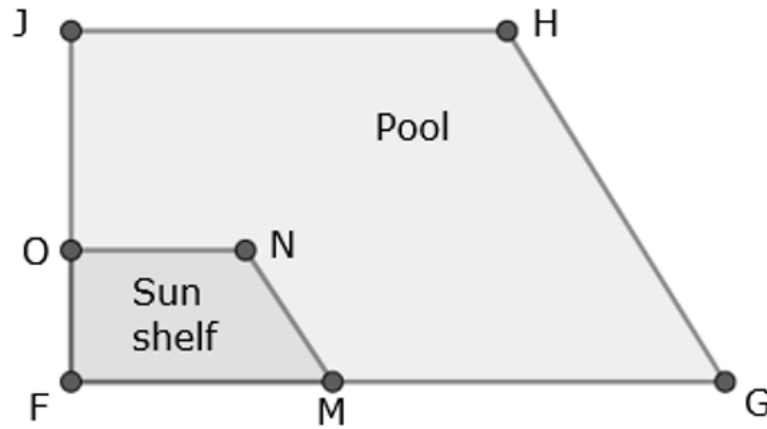
1. What is the value of w ?
2. What is the area of $EFGH$? Include units.
3. What is the perimeter of $JKLM$? Include units.

Tamika created a third similar rectangular frame, $PQRS$, shown below, where $PQ = 6$ in.



4. What is the length of $\overline{SP}$? Include units.
5. What is the perimeter, in inches, of $PQRS$?

Shelley and her family are having a pool built in the backyard. The pool, $FGHJ$, is shaped like a trapezoid, with a smaller similar trapezoid, $FMNO$, for a sun shelf. A diagram of the pool is shown, where $FG = (12k - 6)$ feet (ft), $FM = 10.5$ ft, $NM = 12$ ft, $HG = 24$ ft, $HJ = (12m - 2)$ ft, and $NO = (4m + 3)$ ft.



6. Determine the value of each variable:

$$k = \underline{\hspace{2cm}}$$

$$m = \underline{\hspace{2cm}}$$

7. Determine the value of each side:

$$HJ = \underline{\hspace{2cm}}$$

$$FG = \underline{\hspace{2cm}}$$

8. If $HJ = JF$, what is the perimeter, in feet, of the pool?

Some measurements of angles of $FGHJ$ and $FMNO$ are given.

$$m\angle FMN = (15y + 10)^\circ, m\angle FGH = (18y - 2)^\circ, m\angle OFM = 90^\circ,$$

$$m\angle FJH = (5w + 15)^\circ, m\angle FON = (15w - 15)^\circ$$

9. Determine the value of each variable:

$$y = \underline{\hspace{2cm}}$$

$$w = \underline{\hspace{2cm}}$$

10. What is the measure of $\angle ONM$?